

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

7tLJBQPRtdh3UcG1Ecy2G3B1eSi51m4DciKVovaE8Yvz

Date: 5/1/2025

Compounds: Ibuprofen, Barbiturates

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: IbuBarbiturate

Formula: C18H22N2O4

Weight: 330.38

Drug-likeness: 0.75

Activity:

Analgesic, Sedative

Target Analysis

Cyclooxygenase-2 (COX-2)

Binding Affinity: -8.3

Key target for reducing inflammatory response and pain.

GABAA Receptor

Binding Affinity: -9.2

Important for central nervous system sedation and anxiolytic effects.

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Stomach

Mitigation Suggestions:

- Monitor liver enzymes regularly
- Co-administration with protective agents like proton pump inhibitors

Pharmacokinetics

Absorption:

Moderate oral bioavailability

Distribution:

Widely distributed, crosses blood-brain barrier

Metabolism:

Hepatic via cytochrome P450 enzymes

Excretion:

Renal, with some hepatic elimination

Half-life: 4-6 hours

Detailed Analysis

IbuBarbiturate is a novel compound designed to combine the anti-inflammatory effects of ibuprofen with the sedative properties of barbiturates. Its potential applications include addressing conditions requiring both pain relief and sedation, such as in perioperative settings or chronic pain with concurrent anxiety. The compound operates by simultaneously inhibiting COX enzymes, responsible for prostaglandin synthesis, and potentiating GABAergic neurotransmission, enhancing central nervous system inhibitory effects. Preclinical studies indicate a need for careful dose management due to the risk of liver damage and gastric irritation. The development process involves advanced synthesis techniques and a comprehensive assessment of biological activity and safety profiles to establish its feasibility as a therapeutic option.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

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